8.1 Q8

$$\int \frac{2^{\ln z^3}}{16z}\,dz$$

$\ln z^3 = 3\ln z$

$f(\ln z)$

$u = 3\ln z$

$du = \frac{3}{z}\,dz$

$$= \int \frac{2^{3u}}{48}\,du$$

$$= \frac{2^u}{(\ln 2)48} + C$$

$$= \frac{2^{\ln z}}{(\ln 2)48} + C$$

Given a function of the form

$$\frac{f(\sqrt{x})}{\sqrt{x}} \longrightarrow \text{Let } u = \sqrt{x}$$

$$du = \frac{1}{2\sqrt{x}}\,dx$$

appear in the integrand

$$\downarrow$$

$2f(u)$

Q6

$$\int \frac{dx}{x-\sqrt{x}}$$

$u = x - \sqrt{x}$

$du = 1 - \frac{1}{2\sqrt{x}}\,dx$

$f(y) = \frac{1}{y-1}$

$$= \int \frac{dx}{\sqrt{x}(\sqrt{x}-1)} \rightarrow \frac{f(\sqrt{x})}{\sqrt{x}}$$

$$\frac{x+\sqrt{x}}{x^2-x}$$

$$= \frac{x}{x(x-1)} + \frac{\sqrt{x}}{x^2-x}$$

$$= \int \frac{2}{u-1}\,du$$

$u = \sqrt{x}\,dx$

$du = \frac{1}{2\sqrt{x}}\,dx$

$$= 2\ln|u-1| + C$$

$$= 2\ln|\sqrt{x}-1| + C$$

Q18

$$\int \frac{2^{\sqrt{y}}}{2\sqrt{y}}\,dy$$

$u = \sqrt{y}$

$du = \frac{1}{2\sqrt{y}}\,dy$

$$= \int 2^u\,du$$

$$= \frac{2^u}{\ln 2} + C$$

$$= \frac{2^{\sqrt{y}}}{\ln 2} + C$$

Q26 $\displaystyle\int \frac{6}{\sqrt{y}\,(1+y)}\,dy \rightarrow \frac{f(\sqrt{y})}{\sqrt{y}}$

$$= \int \frac{6\cdot 2}{1+u^2}\,du$$

$$f(x) = \frac{6}{1+x^2}$$

$u = \sqrt{y}$

$du = \frac{1}{2\sqrt{y}}\,dy$

$$= 12\int \frac{1}{1+u^2}\,du$$

$$= 12 \arctan u + C$$

$$= 12 \arctan \sqrt{y} + C$$

Q40 $\displaystyle\int \frac{\sqrt{x}}{1+x^3}\,dx$ Hint: $u = x^{3/2}$.

$$= \int \frac{\sqrt{x}}{1+(x^{\frac{3}{2}})^2}\,dx$$

Idea: If we are given $1+f(x)$ or $1-f(x)$ (usually appears in denom.) Purposely write it as $1 \pm (\sqrt{f(x)})^2$

$$= \frac{2}{3}\int \frac{1}{1+u^2}\,du$$

$u = x^{3/2}$

$du = \frac{3}{2}\sqrt{x}\,dx$

$$= \frac{2}{3}\arctan u + C$$

$$= \frac{2}{3}\arctan x^{3/2} + C$$

Q 36

$$\int \frac{dx}{(2x+1)\sqrt{\underbrace{4x+4x^2}_{1\pm f(x)}}}$$

$$4x^2+4x$$
$$=4(x^2+x)$$
$$=4\left(x^2+x+\left(\tfrac{1}{2}\right)^2-\left(\tfrac{1}{2}\right)^2\right)$$
$$=4\left(\left(x+\tfrac{1}{2}\right)^2-\tfrac{1}{4}\right)$$
$$=4\left(x+\tfrac{1}{2}\right)^2-1$$
$$=(2x+1)^2-1$$

$$=\int \frac{dx}{(2x+1)\sqrt{(2x+1)^2-1}}$$

$u=2x+1$

$du=2\,dx$

$$=\frac{1}{2}\int \frac{du}{u\sqrt{u^2-1}}$$

$$=\frac{1}{2}\,\cancel{\sec u}\,\operatorname{arcsec}|u|+C$$

$$=\frac{1}{2}\operatorname{arcsec}|2x+1|+C$$

Try Q 25 & 27

Q24

$$\int (\sec t + \cot t)^2 \, dt$$

$$= \int \underline{\sec^2 t} + 2\sec t \cdot \cot t + \cot^2 t \quad dt$$

$$\int 2\sec t \cot t \, dt$$

$$= \int 2 \frac{1}{\cos t} \cdot \frac{\cos t}{\sin t} \, dt$$

$$= \int \frac{2}{\sin t} \, dt$$

$$= \int 2 \cancel{\sec}\ \csc t \, dt \qquad \boxed{\csc t}$$

$$= \cancel{2\ln} - 2\ln|\csc t + \cot t| + C$$

$$\int \cot^2 t \, dt \qquad \sin^2 x + \cos^2 x = 1$$

$$1 + \cot^2 x = \csc^2 x$$

$$= \int \csc^2 t - 1 \, dt$$

$$= -\cot t - t + C$$

$\therefore$ Original $= \tan t - 2\ln|\csc t + \cot t|$
$- \cot t - t + C$

$\underline{\sec t}$ $\quad$ $t \sec$ $\quad$ $=$

<u>Der.</u>

$$\frac{d}{dx} \sec x = \sec x \tan x$$

$$\int \sec x \, dx = \ln |\sec x + \tan x|$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\int \tan x \, dx = \ln |\sec x + \sec x|$$

$$= \ln |2 \sec x|$$

$$= \ln |\sec x| + C$$

$$\int \sec^2 x \, dx = \tan x + C$$

$$\int \tan x \sec x \, dx = \sec x + C$$